Vector Algebra

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March 2025

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1 Scalars and Vectors

A **scalar** has magnitude only (for example, mass). A **vector** has magnitude and direction (for example, displacement). A vector in three dimensions is

$$\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}, \qquad \boxed{|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}}.$$

A **unit vector** has magnitude 1. The unit vector along nonzero $\vec{a}$ is $\boxed{\hat{a} = \vec{a}/|\vec{a}|}$. The zero vector has magnitude zero and no defined direction. Equal vectors have equal magnitudes and directions; negative vectors have equal magnitudes and opposite directions.

2 Position and Displacement Vectors

For $P(x, y, z)$, the position vector is $\vec{OP} = x\hat{i} + y\hat{j} + z\hat{k}$. For $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$,

$$\boxed{\vec{AB} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}}.$$

Distance $AB = |\vec{AB}|$.

3 Addition and Scalar Multiplication

For $\vec{a} = (a_1, a_2, a_3)$ and $\vec{b} = (b_1, b_2, b_3)$,

$$\vec{a} + \vec{b} = (a_1 + b_1, a_2 + b_2, a_3 + b_3), \qquad k\vec{a} = (ka_1, ka_2, ka_3).$$

Vector addition follows the triangle and parallelogram laws. Useful properties:

$$\vec{a} + \vec{b} = \vec{b} + \vec{a}, \quad (\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c}), \quad k(\vec{a} + \vec{b}) = k\vec{a} + k\vec{b}.$$

4 Direction Cosines and Direction Ratios

If α, β, γ are the angles a nonzero vector makes with the positive coordinate axes, its direction cosines are

$$l = \cos\alpha = \frac{a_1}{|\vec{a}|}, \quad m = \cos\beta = \frac{a_2}{|\vec{a}|}, \quad n = \cos\gamma = \frac{a_3}{|\vec{a}|}.$$

$$\boxed{l^2 + m^2 + n^2 = 1}.$$

The components (a_1, a_2, a_3) are direction ratios; any proportional triple also gives direction ratios.

5 Section Formula

If P divides the segment from A to B internally in ratio $AP : PB = m : n$, with position vectors $\vec{a}, \vec{b}$, then

$$\boxed{\vec{p} = \frac{n\vec{a} + m\vec{b}}{m + n}}.$$

For the midpoint, $\vec{p} = (\vec{a} + \vec{b})/2$.

6 Dot (Scalar) Product

$$\boxed{\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta = a_1b_1 + a_2b_2 + a_3b_3}.$$

For nonzero vectors,

$$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}, \qquad \vec{a}\perp\vec{b} \iff \vec{a}\cdot\vec{b} = 0.$$

Also $\vec{a}\cdot\vec{a} = |\vec{a}|^2$, and $\vec{a}\cdot\vec{b} = \vec{b}\cdot\vec{a}$. **Example.** For $\vec{a} = (1,2,2)$ and $\vec{b} = (2,0,1)$, $\vec{a}\cdot\vec{b} = 4$, $|\vec{a}| = 3$, $|\vec{b}| = \sqrt{5}$, so $\cos\theta = 4/(3\sqrt{5})$.

7 Cross (Vector) Product

$$\boxed{\vec{a}\times\vec{b} = |\vec{a}||\vec{b}|\sin\theta\,\hat{n}},$$

where $\hat{n}$ is perpendicular to both vectors in the right-hand-rule direction. In component form,

$$\vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}.$$

$$\vec{a}\times\vec{b} = -\vec{b}\times\vec{a}, \qquad \vec{a}\times\vec{a} = \vec{0}.$$

For nonzero vectors, $\vec{a}\parallel\vec{b} \iff \vec{a}\times\vec{b} = \vec{0}$.

8 Geometrical Applications

The area of a parallelogram with adjacent sides $\vec{a},\vec{b}$ is

$$\boxed{A = |\vec{a}\times\vec{b}|}.$$

The area of a triangle with two side vectors $\vec{a},\vec{b}$ from the same vertex is

$$\boxed{A = \frac{1}{2}|\vec{a}\times\vec{b}|}.$$

The scalar projection of $\vec{a}$ along nonzero $\vec{b}$ is $\boxed{(\vec{a}\cdot\vec{b})/|\vec{b}|}$.

9 Useful Unit-Vector Products

$$\hat{i}\cdot\hat{i} = \hat{j}\cdot\hat{j} = \hat{k}\cdot\hat{k} = 1, \qquad \hat{i}\cdot\hat{j} = \hat{j}\cdot\hat{k} = \hat{k}\cdot\hat{i} = 0.$$

$$\hat{i}\times\hat{j} = \hat{k}, \quad \hat{j}\times\hat{k} = \hat{i}, \quad \hat{k}\times\hat{i} = \hat{j}.$$

Reversing the order reverses the sign.